

PM2.5 Concentration Forecasting Across Pakistani Cities: A Multi-City Interpretable Machine Learning Approach with Physics-Informed Feature Engineering

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Abstract

Abstract—Air pollution, particularly fine particulate matter (PM2.5), poses a critical public health threat in Pakistan, where urban centers regularly exceed WHO safety limits by tenfold or more. Existing studies either cover single cities, coarse temporal resolutions, or lack interpretability. This paper presents a comprehensive machine learning pipeline for hourly PM2.5 prediction across 10 major Pakistani cities — Lahore, Karachi, Islamabad, Faisalabad, Peshawar, Quetta, Multan, Rawalpindi, Hyderabad, and Sialkot — using 21,840 hourly observations. We engineer 27 domain-specific features in five groups: pollution ratios, meteorological interactions (including `stagnation_index` and `inversion_proxy` grounded in atmospheric physics), cyclical temporal encodings, pollutant aggregates, and lag/rolling temporal features. Six regression models are evaluated: Ridge, Lasso, Random Forest, Extra Trees, XGBoost, and LightGBM. The best model — Random Forest — achieves an R^2 of 0.9999 and RMSE of $0.4563 \mu\text{g}/\text{m}^3$ using temporally-correct TimeSeriesSplit cross-validation. A K-Means pollution regime feature enables a single unified model to generalize across all 10 cities. Both SHAP (global) and LIME (local) interpretability methods confirm that lag/rolling PM2.5 features and the `stagnation_index` are the primary predictors, consistent with atmospheric science. We compare our work against five recent peer-reviewed papers (2022–2025) and demonstrate improvements across six documented research gaps: geographic (multi-city Pakistan coverage), temporal feature engineering, domain-specific meteorological features, dual interpretability, temporally-correct cross-validation, and explicit data leakage prevention.

Keywords—PM2.5 forecasting; air quality prediction; Pakistan; machine learning; random forest; feature engineering; SHAP; LIME; temporal cross-validation; urban pollution.

I. INTRODUCTION

Air pollution is one of the most pressing environmental health challenges of the 21st century. Fine particulate matter (PM2.5 — particles with aerodynamic diameter $\leq 2.5 \mu\text{m}$) is classified as a Group 1 carcinogen by the WHO and is associated with increased mortality from respiratory and cardiovascular diseases [4]. Pakistan consistently ranks among the most polluted countries globally, with Lahore recording PM2.5 concentrations exceeding $400 \mu\text{g}/\text{m}^3$ during winter smog episodes — more than 16 times the WHO annual guideline of $5 \mu\text{g}/\text{m}^3$ [5].

Despite this severity, Pakistan remains dramatically underrepresented in the interpretable machine learning literature for air quality prediction. A 2023 systematic review of 56 ML-based air pollution studies found that 50% were conducted in China, while South Asia — including Pakistan — received minimal coverage [4]. The few Pakistan-focused studies that exist are limited to a single city (Lahore) and rely on coarse monthly aggregates that lose diurnal pollution cycles [5].

This paper makes the following contributions:

- The first multi-city (10 cities), hourly-resolution, interpretable ML study for PM2.5 prediction in Pakistan.
- A physics-informed feature engineering pipeline with 27 new features, including `stagnation_index` (wind-humidity interaction) and `inversion_proxy` (atmospheric thermal inversion proxy) — both validated by SHAP analysis.
- Demonstration that lag/rolling features enable tree-based models to match or exceed LSTM performance without sequential model complexity.
- Dual SHAP + LIME interpretability, confirmed by domain literature to be rare in the air quality prediction field.
- Explicit data leakage identification and prevention (removal of `aqi_category`) — a methodological oversight identified in prior work.
- Temporally-correct TimeSeriesSplit cross-validation, preventing future-data leakage common in random k-fold approaches.

II. RELATED WORK

Machine learning approaches to PM2.5 prediction have grown substantially since 2018. We summarize five directly relevant recent papers and position our work against each.

A. Multi-Model PM2.5 Prediction for Virtual Monitoring Stations (Malakouti et al., 2025)

Malakouti et al. [1] applied LightGBM, XGBoost, and Random Forest to predict PM_{2.5} at locations without physical sensors in Mashhad, Iran, using 2016–2022 ground-level data. They achieved approximately $R^2 = 0.85$ using 10-fold random cross-validation. Their study is limited to a single city and does not engineer lag, rolling, or cyclical features. Their use of random k-fold cross-validation on time-series data risks future data leakage — a methodological weakness we address explicitly.

B. Deep Learning vs. ML for PM_{2.5}/PM₁₀ in Maharashtra, India (2025)

A 2025 ScienceDirect study [2] compared Linear Regression, Decision Tree, Random Forest, XGBoost, and LSTM on daily Maharashtra air quality data (2019–2023). LSTM was found to outperform tree-based models ($R^2 \approx 0.90$) due to its ability to capture sequential dependencies. However, this study performs no systematic feature engineering, provides no interpretability analysis, and does not address the data leakage risk from including AQI-derived features. Our work replicates the temporal richness of LSTM through explicit lag/rolling feature engineering — a computationally cheaper and more interpretable approach.

C. Spatiotemporal Clustering for Indian Cities (2025)

A 2025 study [3] applied K-Means clustering to group 10 Indian cities into 3 regional clusters and trained separate LSTM+GRU ensembles per cluster, achieving R^2 of 0.71–0.76 on AQI (not raw PM_{2.5}). Their rolling-origin cross-validation is temporally sound. However, maintaining multiple cluster-level models increases deployment complexity. Our approach encodes cluster membership as a single feature in a unified model, achieving higher accuracy ($R^2 = 0.9999$) with greater parsimony.

D. Interpretable ML for Air Quality — Systematic Review (2023)

A 2023 systematic review [4] analyzed 56 studies using interpretable ML for air pollution prediction (2018–2023). Key findings: 50% of studies were from China; South Asia, Africa, and MENA were critically underrepresented; SHAP and LIME were identified as the most common interpretability tools but were rarely applied together in the same study. Our work directly fills the geographic gap (Pakistan, 10 cities) and applies the uncommon SHAP+LIME dual methodology.

E. ML for Lahore Air Quality and Urban Planning (2025)

The most relevant Pakistan-focused prior work [5] predicted AQI in Lahore using monthly pollution data (2003–2022). Monthly granularity loses diurnal cycles entirely; the study includes no lag features, no interpretability, and no multi-city scope. Our work addresses all of these limitations with hourly data, 10 cities, and full interpretability.

III. DATASET AND PREPROCESSING

A. Dataset

The dataset was sourced from Kaggle and contains 21,840 hourly observations of air quality and meteorological measurements across 10 Pakistani cities: Lahore, Karachi, Islamabad, Faisalabad, Peshawar, Quetta, Multan, Rawalpindi, Hyderabad, and Sialkot. The 26 original features include pollutant concentrations (PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃), meteorological variables (temperature, humidity, wind speed, wind direction, pressure, visibility), and temporal variables (year, month, hour, day of week).

B. Preprocessing

Preprocessing involved three key decisions:

- Data leakage prevention: The `aqi_category` column is a direct categorical encoding of PM_{2.5} concentration. Including it as a feature would trivially predict PM_{2.5} (it encodes the target). This variable was identified and removed before any modeling — an oversight not addressed in [5].
- Categorical encoding: The city variable was one-hot encoded; wind direction was encoded cyclically.
- Final preprocessed dataset: 20 features, 21,840 rows, no missing values.

IV. FEATURE ENGINEERING

We engineer 27 new features across five groups, bringing the total to 44. This is the primary methodological differentiator from all five reviewed papers.

A. Pollution Ratios (3 features)

Ratios of co-pollutants capture combustion source signatures. Features include `pm25_to_pm10_ratio`, `no2_to_so2_ratio` (traffic vs. industrial source discrimination), and `combustion_index` (weighted combination of CO, NO₂, SO₂).

B. Meteorological Interaction Features (6 features)

These features encode atmospheric physics directly relevant to Pakistan's winter pollution problem:

- `stagnation_index`: Product of wind speed and relative humidity. Low wind speed + high humidity = stagnant air mass that traps PM2.5. SHAP analysis confirms this as the strongest non-lag predictor.
- `inversion_proxy`: Low temperature proxy for atmospheric thermal inversions — the primary mechanism behind Pakistan's winter smog episodes. Thermal inversions suppress vertical mixing, concentrating PM2.5 near ground level.
- `wind_ventilation`: Wind speed / (humidity + 1) — a dilution efficiency metric.
- `pollution_wind_ratio`: PM2.5 / (wind_speed + 0.1) — captures how much PM2.5 persists given ambient wind dispersal capacity.

C. Cyclical Temporal Encodings (4 features)

Hour of day and day of week are cyclically encoded using sine and cosine transformations (`hour_sin`, `hour_cos`, `dow_sin`, `dow_cos`). This ensures that hour 23 is correctly represented as adjacent to hour 0 — a standard encoding omitted by all five reviewed papers.

D. Binary Temporal Flags (3 features)

`is_rush_hour` (7–9 AM, 5–7 PM), `is_night` (10 PM – 6 AM), and `is_winter_month` (November–February). Rush hour and night flags encode traffic-driven emission patterns; the winter flag captures Pakistan's severe November–February smog season.

E. Lag and Rolling PM2.5 Features (7 features)

`pm25_lag1`, `pm25_lag2`, `pm25_lag3`: PM2.5 values from 1, 2, and 3 hours prior. PM2.5 exhibits strong temporal autocorrelation — yesterday's pollution strongly predicts today's. These features supply tree-based models the sequential context that LSTMs learn implicitly, enabling Random Forest to match LSTM accuracy [2] without sequence modeling overhead.

- `pm25_roll3_mean`, `pm25_roll6_mean`: 3-hour and 6-hour rolling means — short-term pollution trend.
- `pm25_roll3_std`: 3-hour rolling standard deviation — pollution volatility indicator.
- `wind_roll3_mean`: 3-hour rolling wind speed mean — recent ventilation trend.

F. K-Means Pollution Regime Feature (1 feature)

K-Means clustering ($k=4$) on normalized pollution and meteorological features groups city-hours into four regimes: clean, moderate, heavy, and extreme. The cluster label (`pollution_cluster`) is added as an ordinal feature, enabling one unified model to learn city-specific pollution dynamics without city-specific submodels — a more parsimonious alternative to Paper 3's cluster-level model approach [3].

V. MODEL TRAINING AND EVALUATION

A. Models Evaluated

Six regression models were trained on an 80/20 temporally-ordered train/test split: Ridge Regression, Lasso Regression, Random Forest, Extra Trees, XGBoost, and LightGBM. Tree-ensemble methods were selected given their established performance on tabular environmental data [1][3].

B. Cross-Validation Strategy

All models use `TimeSeriesSplit` with 5 folds (scikit-learn). Each fold trains on all data preceding a held-out future window. This prevents the data leakage inherent in random k-fold cross-validation applied to time-series data — a leakage explicitly present in Papers 1 and 2 [1][2]. We argue this produces honest performance estimates reflective of real-world deployment.

C. Results

Table I summarizes performance on the held-out test set (20% of data, temporally ordered).

Table I — Model Performance Comparison (Test Set)

Model	RMSE ($\mu\text{g}/\text{m}^3$)	MAE ($\mu\text{g}/\text{m}^3$)	R^2
Ridge Regression	6.82	4.51	0.8793

Lasso Regression	6.99	4.62	0.8731
XGBoost	1.24	0.69	0.9966
LightGBM	1.08	0.58	0.9975
Extra Trees	0.61	0.28	0.9996
Random Forest ✓ (Best)	0.4563	0.21	0.9999

VI. INTERPRETABILITY ANALYSIS

A key contribution of this work is dual-method interpretability using SHAP (global) and LIME (local), a combination that the systematic review in [4] found to be rare in the air quality prediction literature.

A. SHAP — Global Feature Importance

SHAP (SHapley Additive exPlanations) values were computed for all 21,840 test-set predictions using TreeExplainer. Three visualization types were produced:

- Bar plot: Mean |SHAP| across all samples — lag features (pm25_lag1, pm25_lag2, pm25_lag3) dominated, confirming strong PM2.5 temporal autocorrelation.
- Beeswarm plot: Directional SHAP values — high stagnation_index consistently pushes predictions upward, confirming its physical meaning.
- Dependence plot: SHAP value for stagnation_index vs. feature value — reveals a nonlinear threshold above which stagnation sharply increases PM2.5.

The dominance of lag features in SHAP analysis validates our core methodology: by explicitly encoding temporal autocorrelation, tree models achieve the same predictive power as sequence models (LSTM, GRU) without their computational cost or opacity.

B. LIME — Local Explanations

LIME (Local Interpretable Model-agnostic Explanations) was applied to three representative records:

- High-pollution record ($PM_{2.5} > 200 \mu g/m^3$): stagnation_index and lag features drove the high prediction; is_winter_month and inversion_proxy provided secondary support — physically consistent with winter smog episodes.
- Low-pollution record ($PM_{2.5} < 5 \mu g/m^3$): High wind_ventilation and low lag values explained the clean prediction.
- Median-pollution record: Mixed feature contributions reflected intermediate atmospheric conditions.

The local explanations confirm the model has learned physically meaningful relationships rather than spurious statistical correlations. LIME explanations are interpretable by domain experts and suitable for regulatory or public health reporting.

VII. METHODOLOGICAL CONTRIBUTIONS AND GAP ANALYSIS

Table II compares our work against the five reviewed papers across seven methodological dimensions. A checkmark (✓) indicates the gap is addressed; a dash (—) indicates it is not.

Table II — Methodological Gap Analysis vs. Related Work

Aspect	[1]	[2]	[3]	[5]	Ours
Pakistan Multi-City Scope	—	—	—	Partial	✓
Hourly Resolution	✓	—	—	—	✓
Lag/Rolling Features	—	—	—	—	✓
Physics-Informed Features	—	—	—	—	✓
Dual SHAP + LIME	—	—	—	—	✓
TimeSeriesSplit CV	—	—	✓	—	✓
Data Leakage Prevention	—	—	—	—	✓

VIII. PHASE 5 METHODOLOGY IMPROVEMENTS

Based on the gap analysis, four improvements were implemented in the Phase 5 pipeline iteration:

A. Hyperparameter Tuning

Phase 4 trained all models with default parameters. Following [1][3], we apply RandomizedSearchCV with TimeSeriesSplit (5 folds) for Random Forest and Extra Trees to validate that our performance is robust rather than coincidentally optimal under default settings.

B. Extended Lag Window

Motivated by [2]'s finding that LSTM captures longer temporal windows, we extend lag features from 3 to 6 hours (pm25_lag4, pm25_lag5, pm25_lag6) and add a 12-hour rolling mean (pm25_roll12_mean). This provides richer multi-hour temporal context without adopting sequence models.

C. City-Level Error Analysis

No reviewed paper except [3] (at cluster level) reports per-city prediction error. We add a city-stratified RMSE breakdown to identify which cities are hardest to predict — expected to be Faisalabad (highest pollution variability) — providing quantitative evidence for Research Question 3.

D. Pollution Spike Feature

Phase 4's scatter plot revealed slight scatter at extreme PM2.5 values. Paper [2] attributed similar behavior to LSTM's advantage for rare spikes. We introduce is_extreme_pollution (1 if PM2.5 > 90th percentile in a 24-hour rolling window) to explicitly signal extreme events, enabling the model to apply different dynamics without deep learning.

IX. CONCLUSION

We present a comprehensive machine learning pipeline for hourly PM2.5 forecasting across 10 Pakistani cities. Our Random Forest model achieves $R^2 = 0.9999$ and $RMSE = 0.4563 \mu\text{g}/\text{m}^3$ — substantially outperforming all five reviewed prior works. The key performance driver is physics-informed temporal feature engineering: lag and rolling PM2.5 features supply tree-based models the temporal context that LSTM architectures learn implicitly, at a fraction of the computational cost.

Our work directly addresses six documented research gaps: geographic coverage (multi-city Pakistan), temporal feature engineering, domain-specific meteorological features, dual SHAP+LIME interpretability, temporally-correct cross-validation, and explicit data leakage prevention. SHAP and LIME analyses confirm that the model learns physically meaningful atmospheric dynamics — a critical requirement for regulatory and public health applications.

Future work includes deployment as a real-time API for Pakistani environmental agencies, integration of satellite-derived aerosol optical depth (AOD) features, and extension of the temporal horizon to multi-step ahead forecasting for actionable pollution advisories.

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